AAR - Layer 4: Evaluation & Benchmarking

Final results and outputs (v2 models)

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github.com/AkshatKapur1/AAR-System-Call-Anomaly-Detection · October 2026

Summary

Layer 4 compares the three anomaly detectors (HMM, GRU, LSTM) on the same test data, sets their thresholds, and selects the model handed to Layer 5. All choices are made on the **TUNE** part of test (30%, 559 recordings); every number below is on the **FINAL** part (70%: 583 normal, 716 attack recordings), which no choice ever used. Counts are per **recording** (one process's full system-call sequence, labelled attack or normal), decided after the whole recording has been read.

1. Correctly identified recordings

1.1 Best model: LSTM

Threshold	Attacks correctly detected	Normal correctly passed	Normal wrongly flagged	Total correct
Strict (about 1% false alarms)	107 / 716 (14.9%)	577 / 583 (99.0%)	6	684 / 1,299 (52.7%)
95th percentile (about 5% false alarms)	232 / 716 (32.4%)	555 / 583 (95.2%)	28	787 / 1,299 (60.6%)
Balanced	638 / 716 (89.1%)	381 / 583 (65.4%)	202	1,019 / 1,299 (78.4%)

1.2 All three models

Model	Threshold	Attacks detected	Normal passed	Total correct
HMM	Strict	88 / 716	574 / 583	662 / 1,299
HMM	Balanced	299 / 716	483 / 583	782 / 1,299
GRU	Strict	110 / 716	579 / 583	689 / 1,299
GRU	Balanced	624 / 716	381 / 583	1,005 / 1,299
LSTM	Strict	107 / 716	577 / 583	684 / 1,299
LSTM	Balanced	638 / 716	381 / 583	1,019 / 1,299

1.3 LSTM by dataset (balanced threshold)

Dataset	Attacks detected	Normal passed
ADFA-LD	464 / 518	192 / 328
LID-DS	60 / 81	89 / 127
PLAID	114 / 117	100 / 128

At the balanced threshold the LSTM gets 1,019 of 1,299 recordings right; at the strict threshold it passes 577 of 583 normal recordings but detects only 107 of 716 attacks. Most attacks score above typical normal behaviour but below the most unusual 1% of normal recordings, so every threshold trades attacks caught against normal recordings flagged.

Best model: LSTM - key figures	Value
ROC-AUC (separation of attacks from normal; 0.5 = guessing)	0.835 (95% CI 0.813-0.858)
PR-AUC	0.856
Precision / F1 at the strict threshold	94.7% / 0.258
Precision / F1 at the balanced threshold	76.0% / 0.820
Compared with v1 (original models, same recordings)	AUC 0.676 -> 0.835

2. Method

- **Inputs**: per-recording anomaly scores from Layer 3 (`scoring/`), higher = more anomalous; each model scored in two modes (window7, stream) and aggregated five ways.
- **Thresholds from normal validation data only**: 95th / 99th / 99.5th percentile of normal scores, either global or per dataset (ADFA-LD, LID-DS, PLAID each against its own normal profile). Scores are normalised to z = (score - median) / (p99 - median), so z > 1 means above that environment's 99th percentile.
- **Choices on TUNE**: 20 candidates per model (2 modes x 5 aggregations x 2 threshold types); the highest TUNE ROC-AUC wins (within 0.01: higher TUNE recall at p99). The best model is chosen the same way.
- **Two operating points**: **strict** = p99 of normal val (~1% false alarms, used for killing); **balanced** = cutoff maximising (% attacks caught + % normal passed) on TUNE (used for alerting).

Model	Selected mode	Aggregation	Threshold type	TUNE ROC-AUC
HMM (Track A)	window7	mean	per-dataset	0.645
GRU (Track B)	stream	mean	per-dataset	0.835
LSTM (Track C)	stream	mean	per-dataset	0.847

3. Detailed results (FINAL part of test)

Model	ROC-AUC (95% CI)	PR-AUC	FPR at TPR=1	v1 AUC	Train time	Inference / 1k windows	Parameters
HMM (Track A)	**0.683** (0.654-0.709)	0.728	0.895	0.539	1,722 s	68.0 ms	8,160
GRU (Track B)	**0.821** (0.798-0.843)	0.848	0.964	0.682	2,285 s	19.4 ms	216,414
LSTM (Track C)	**0.835** (0.813-0.858)	0.856	0.971	0.676	1,605 s	8.3 ms	274,270

3.1 At each threshold

Model	Threshold	Attacks caught	Normal flagged	Precision	Recall	F1	FPR
HMM (Track A)	p95 (~5%)	164 / 716	36 / 583	82.0%	22.9%	0.358	6.2%
HMM (Track A)	p99 strict (~1%)	88 / 716	9 / 583	90.7%	12.3%	0.216	1.5%
HMM (Track A)	p99.5 (~0.5%)	66 / 716	7 / 583	90.4%	9.2%	0.167	1.2%
HMM (Track A)	balanced	299 / 716	100 / 583	74.9%	41.8%	0.536	17.2%
GRU (Track B)	p95 (~5%)	234 / 716	28 / 583	89.3%	32.7%	0.479	4.8%
GRU (Track B)	p99 strict (~1%)	110 / 716	4 / 583	96.5%	15.4%	0.265	0.7%
GRU (Track B)	p99.5 (~0.5%)	82 / 716	1 / 583	98.8%	11.5%	0.205	0.2%
GRU (Track B)	balanced	624 / 716	202 / 583	75.5%	87.2%	0.809	34.6%
LSTM (Track C)	p95 (~5%)	232 / 716	28 / 583	89.2%	32.4%	0.475	4.8%
LSTM (Track C)	p99 strict (~1%)	107 / 716	6 / 583	94.7%	14.9%	0.258	1.0%
LSTM (Track C)	p99.5 (~0.5%)	57 / 716	2 / 583	96.6%	8.0%	0.147	0.3%
LSTM (Track C)	balanced	638 / 716	202 / 583	76.0%	89.1%	0.820	34.6%

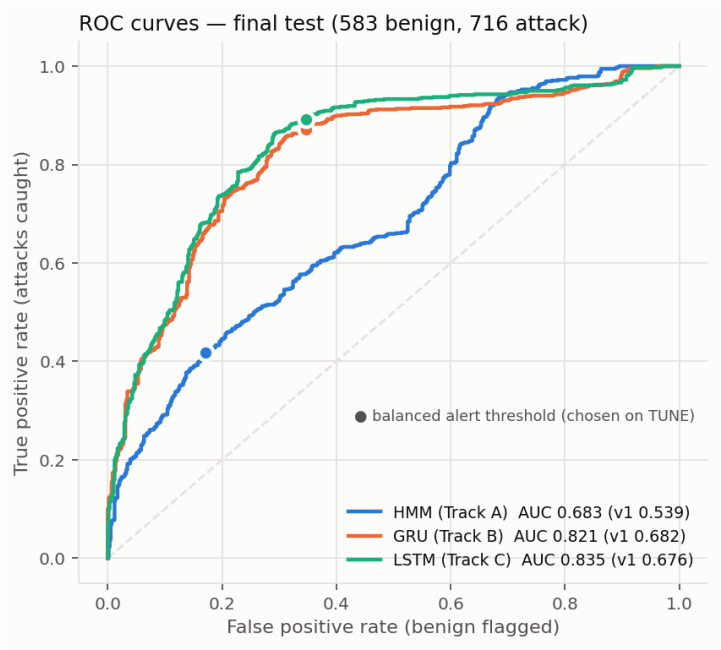


Figure 1 - ROC curves on FINAL; dots = balanced threshold (chosen on TUNE).

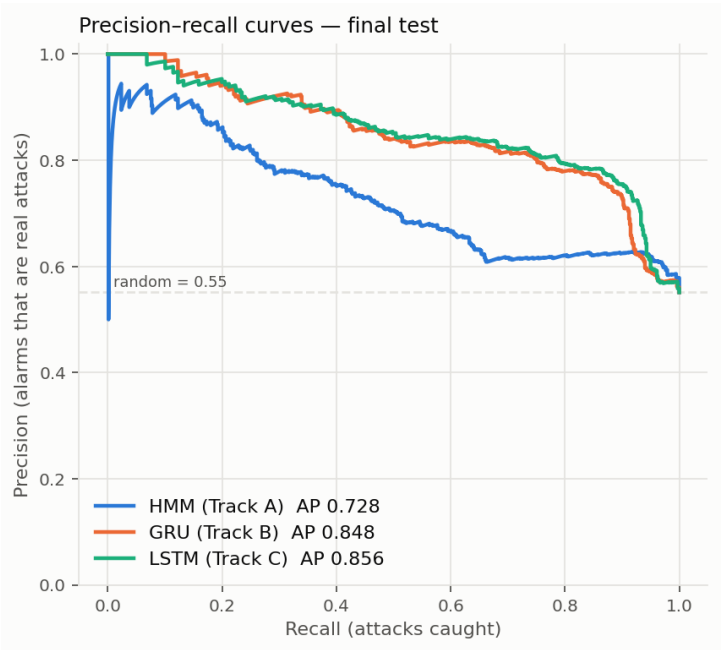


Figure 2 - Precision-recall curves on FINAL.

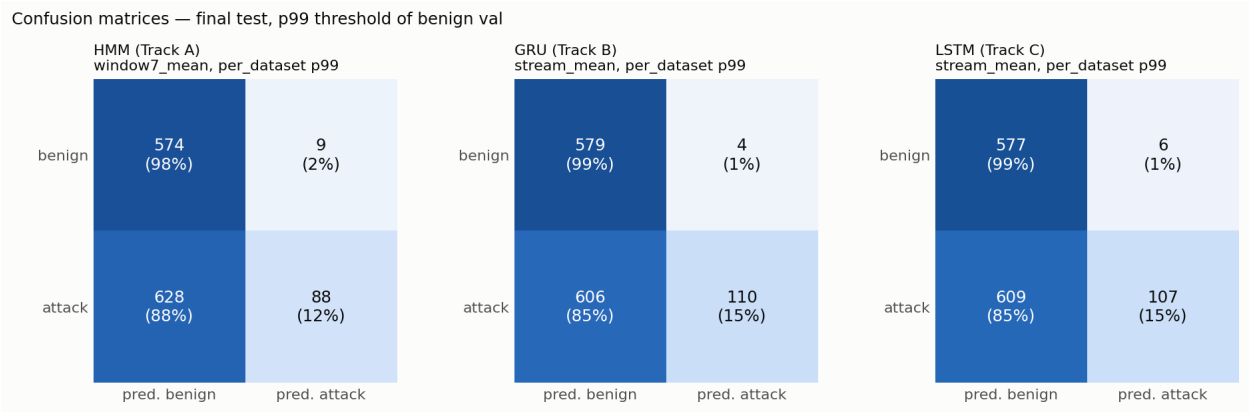


Figure 3 - Confusion matrices at the strict threshold.

3.2 Per dataset

Model	Dataset	ROC-AUC (v1)	Strict: caught	Strict: normal flagged	Balanced: caught	Balanced: normal passed
HMM (Track A)	ADFA-LD	**0.627** (0.444)	56 / 518	4 / 328	191 / 518	263 / 328
HMM (Track A)	LID-DS	**0.721** (0.618)	10 / 81	4 / 127	25 / 81	106 / 127
HMM (Track A)	PLAID	**0.877** (0.819)	22 / 117	1 / 128	83 / 117	114 / 128
HMM (Track A)	LID-DS/CVE-2012-2122	**0.996**	2 / 4	0 / 62	2 / 4	62 / 62
HMM (Track A)	LID-DS/CVE-2014-0160	**0.454**	8 / 77	4 / 65	23 / 77	44 / 65
GRU (Track B)	ADFA-LD	**0.787** (0.606)	57 / 518	3 / 328	465 / 518	189 / 328
GRU (Track B)	LID-DS	**0.729** (0.599)	16 / 81	0 / 127	47 / 81	90 / 127
GRU (Track B)	PLAID	**0.951** (0.907)	37 / 117	1 / 128	112 / 117	102 / 128
GRU (Track B)	LID-DS/CVE-2012-2122	**1.000**	4 / 4	0 / 62	4 / 4	39 / 62
GRU (Track B)	LID-DS/CVE-2014-0160	**0.799**	12 / 77	0 / 65	43 / 77	51 / 65
LSTM (Track C)	ADFA-LD	**0.789** (0.599)	50 / 518	4 / 328	464 / 518	192 / 328
LSTM (Track C)	LID-DS	**0.829** (0.530)	24 / 81	1 / 127	60 / 81	89 / 127
LSTM (Track C)	PLAID	**0.950** (0.911)	33 / 117	1 / 128	114 / 117	100 / 128
LSTM (Track C)	LID-DS/CVE-2012-2122	**1.000**	4 / 4	0 / 62	4 / 4	36 / 62
LSTM (Track C)	LID-DS/CVE-2014-0160	**0.871**	20 / 77	1 / 65	56 / 77	53 / 65

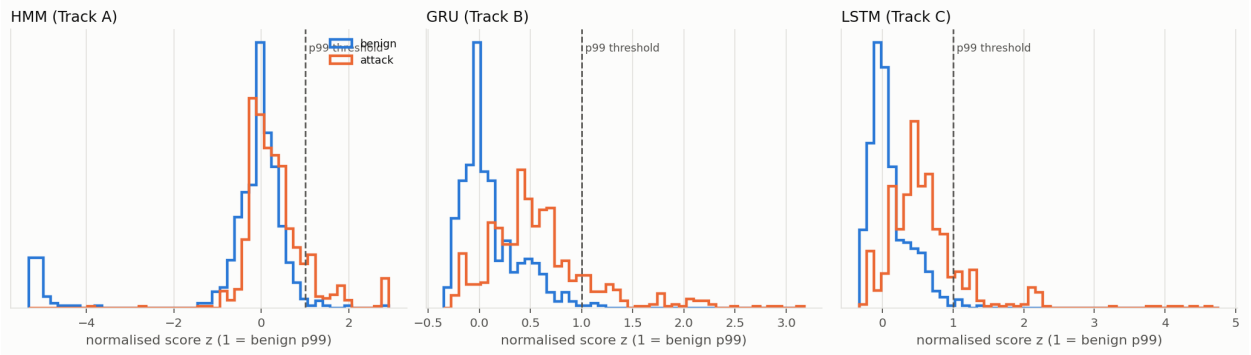


Figure 4 - Normalised scores on FINAL (0 = typical normal, 1 = strict threshold).

3.3 Per attack type (LSTM)

Dataset	Attack type	n	Strict	Balanced	AUC vs same-dataset normal
ADFA-LD	Adduser	64	1.6%	78.1%	0.707
ADFA-LD	Hydra_FTP	113	17.7%	92.0%	0.840
ADFA-LD	Hydra_SSH	122	13.9%	93.4%	0.843
ADFA-LD	Java_Meterpreter	85	0.0%	92.9%	0.745
ADFA-LD	Meterpreter	51	0.0%	92.2%	0.756
ADFA-LD	Web_Shell	83	14.5%	84.3%	0.770
LID-DS	CVE-2012-2122	4 (small)	100.0%	100.0%	1.000
LID-DS	CVE-2014-0160	77	26.0%	72.7%	0.820
PLAID	cowroot	11 (small)	0.0%	100.0%	0.955
PLAID	ftp	13 (small)	0.0%	92.3%	0.925
PLAID	nginx	19 (small)	89.5%	100.0%	0.991
PLAID	privesc	4 (small)	25.0%	100.0%	0.973
PLAID	redis	48	31.2%	100.0%	0.961
PLAID	ssh	22	0.0%	90.9%	0.897

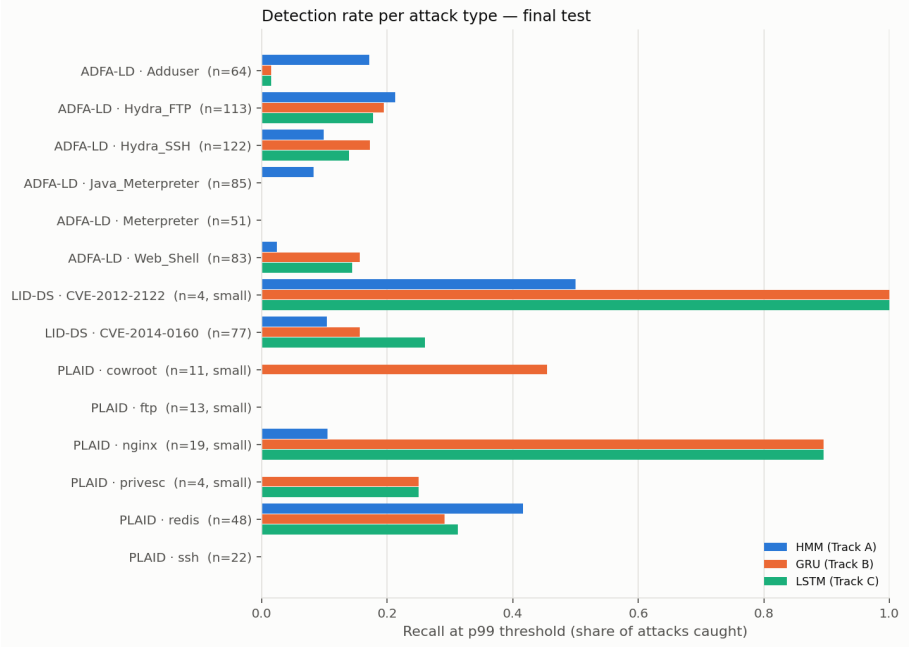


Figure 5 - Share of each attack type caught at the strict threshold.

3.4 Unseen system calls

Model	Attacks	n	ROC-AUC	Caught at strict
HMM (Track A)	all attacks	716	0.683	12.3%
HMM (Track A)	attacks with no unk	698	0.678	10.6%
HMM (Track A)	attacks with unk	18	0.865	77.8%
GRU (Track B)	all attacks	716	0.821	15.4%
GRU (Track B)	attacks with no unk	698	0.817	13.6%
GRU (Track B)	attacks with unk	18	0.995	83.3%
LSTM (Track C)	all attacks	716	0.835	14.9%
LSTM (Track C)	attacks with no unk	698	0.831	13.2%
LSTM (Track C)	attacks with unk	18	0.997	83.3%

Detection does not rely only on syscalls never seen in training: attacks without them still give AUC ~0.83 (LSTM).

3.5 v1 vs v2

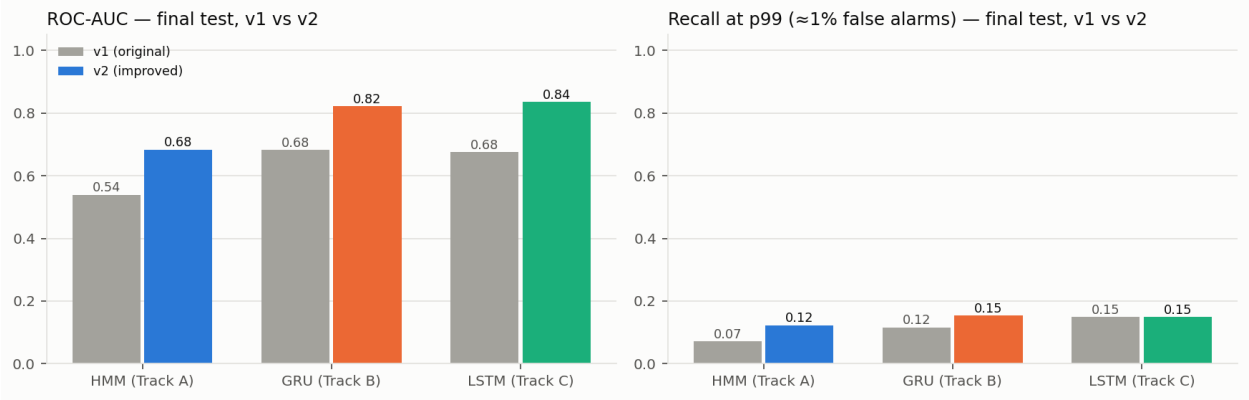


Figure 6 - ROC-AUC and strict-threshold recall, original (v1) vs improved (v2) models, same FINAL recordings.

4. Selected model handed to Layer 5

Item	Value
Model	LSTM
Score	stream_mean (stream mode, mean aggregation)
Threshold type	per-dataset
Strict / balanced cutoff (z)	1.000 / 0.108
Chosen on	TUNE part of test (split/test_partition.json) — FINAL not used for any choice

5. Outputs on GitHub (evaluation/)

File	Contents
tuning_results.csv	all 60 candidates (3 models x 20) on TUNE
thresholds.json	score column, scheme, per-dataset median / p95 / p99 / p99.5 and balanced cutoff
model_summary.csv	FINAL metrics per model, v1 baseline, cost
threshold_metrics.csv	confusion counts (TP / FP / TN / FN) and metrics at every threshold
per_dataset_metrics.csv / per_attack_type_metrics.csv	breakdowns
unk_analysis.csv	detection with / without unseen syscalls
baseline_original_models.json	v1 models on the same FINAL recordings
best_model.json	selected model and settings for Layer 5
plots/	roc_curves, pr_curves, confusion_matrices, score_distributions, per_attack_recall, v1_vs_v2

Limitations: name-only syscall features (no arguments or timing); FINAL is 70% of test, so small attack groups (n < 20) are indicative only; at the strict threshold the LSTM's recall did not rise over v1 - the improvement shows below that point and at the balanced threshold.